

MATH 1300 D : Differential Calculus with applications

TEST 1

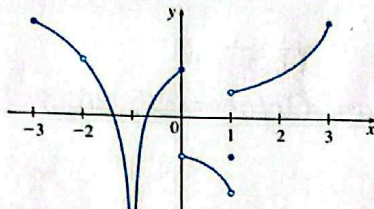
Sunday, October 26, 2025

Solutions

- This test contains 8 pages and 18 questions (10 MCQ (Multiple Choice) in Part A, 5 Short answers in Part B and 3 long answers in Part C plus a bonus question.
- This is a 90-minute test. Use your time sensibly !
- This is a closed book exam. Lecture slides, notes, books, textbook, online resources are definitely NOT allowed.
- Calculators are NOT allowed!

Part A: Multiple Choice Questions (1 mark each). Shade the bubble corresponding to your answer on the MCQ answer log sheet at the back.

1. Consider the graph of a function f below:



Select the FALSE statement below:

- (A) $f(1)$ is defined. (B) f has infinite discontinuity at -1 .
 (C) $\lim_{x \rightarrow -2} f(x)$ exists. (D) f is continuous from right at $x = 0$.
 (E) $\lim_{x \rightarrow 0^-} f(x) = f(0)$

$$\lim_{x \rightarrow 0^+} f(x) \neq f(0)$$

2. TRUE or FALSE. The *Intermediate Value Theorem* tells us that there is a solution of the equation $x^2 + 2 = 5\sqrt{x}$, on the interval $(0, 1)$.

- (A) TRUE (B) FALSE

$$f(x) = x^2 + 2 - 5\sqrt{x} = 0$$

$$f(0) = 2 > 0 \quad f(1) = -2 < 0$$

3. Given that $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 0$, $\lim_{x \rightarrow c} h(x) = 0$

Select the TRUE statement below:

- (A) $\lim_{x \rightarrow c} f(x)g(x)$ does not exist. (B) $\lim_{x \rightarrow c} \frac{g(x)}{h(x)}$ may exist.
 (C) $\lim_{x \rightarrow c} \frac{h(x)}{f(x)}$ does not exist. (D) $\lim_{x \rightarrow c} h(x)g(x)$ does not exist.

4. Let f be some function for which you know that:

for all $\epsilon > 0$ there is a $\delta > 0$ so that, if $0 < |x - 1| < \delta$ then $|f(x) - 5| < \epsilon$

Select the FALSE statement below:

- (A) $\lim_{x \rightarrow 1} f(x) = 5$
 (B) for $\delta^* \leq \delta$, $0 < |x - 1| < \delta^*$ implies $|f(x) - 5| < \epsilon$
 (C) $|x - 1| < \delta$ implies $|f(x) - 5| < \epsilon$
 (D) If $1 < x < 1 + \delta$ then $|f(x) - 5| < \epsilon$.
 (E) If $1 - \delta < x < 1$ then $|f(x) - 5| < \epsilon$.

$$x \neq 1$$

$$x = 1, \text{ so } |f(1) - 5| < \epsilon \text{ not true}$$

5. TRUE or FALSE. If f and g are differentiable at x and $g(x) \neq 0$ then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f(x)g'(x) - g(x)f'(x)}{[g(x)]^2} \leftarrow \text{should be other way round}$$

(A) TRUE

(B) FALSE

6. Consider the function :

$$f(x) = \begin{cases} -x^2 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \frac{1}{x^2} & \text{if } x > 0 \end{cases}$$

Select the INCORRECT statement below:

- (A) f is continuous at $x = 2$.
 (B) f is continuous from left at $x = 0$.
 (C) f has an infinite discontinuity at $x = 0$.
 (D) f is continuous from right at $x = 0$.

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(\frac{1}{x^2} \right) \neq 0 = f(0)$$

7. Consider $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx}$, where a, b non-zero constants and $a \neq b$.

$$\frac{\sin(ax)}{bx} = \frac{a \sin ax}{b ax}$$

Select the TRUE statement below:

- (A) $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = 1$ (B) $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$ (C) $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = \frac{b}{a}$
 (D) $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = ab$ (E) $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx}$ does not exist.

8. TRUE or FALSE. If $f(x)$ is not continuous at $x = a$ then $f(x)$ is not differentiable at $x = a$.

(A) TRUE

(B) FALSE

9. Select from below the CORRECT derivative of $\tan^{-1}(\sqrt{x})$:

$$\frac{d}{dx} \tan^{-1} \sqrt{x} = \frac{1}{1+(\sqrt{x})^2} \cdot \frac{d}{dx} \sqrt{x}$$

- (A) $\frac{1}{2\sqrt{x}(1+x^2)}$, (B) $\frac{1}{1+x^2}$, (C) $\frac{1}{x(1+x^2)}$, (D) $\frac{1}{2\sqrt{x}(1+x)}$

10. TRUE or FALSE, The equation $e^{\ln x} = x$ is valid for all real values of x .

(A) TRUE,

(B) FALSE

valid for only positive values of x
 since $\ln x$ is defined for $x > 0$

Part B: Short Answer. Write only your final answer in the box provided. No need to show your work in this section.

1. (2 marks) Solve the inequality $0 < |x - 2| < 3$. Give your answer in interval form.

Answer

$$x \text{ in } (-1, 2) \cup (2, 5) \\ \text{OR } (-1, 5) \setminus \{2\}$$

2. (2 marks) Decide on intuitive grounds whether or not the following limit exists. If the limit exists, state the limit below. If the limit does not exist, write "does not exist".

$$\frac{\sqrt{x}-3}{x-9} = \frac{\sqrt{x}-3}{(\sqrt{x}-3)(\sqrt{x}+3)}$$

$$\lim_{x \rightarrow 9} \frac{\sqrt{x}-3}{x-9}$$

Answer

$$\frac{1}{6}$$

3. (2 marks) Decide on intuitive grounds whether or not the following limit exists. If the limit exists, state the limit below. If the limit does not exist, write "does not exist".

$$\frac{\sqrt{x}+3}{x-9} = \frac{\frac{1}{\sqrt{x}} + \frac{3}{x}}{1 - \frac{9}{x}}$$

$$\lim_{x \rightarrow \infty} \dots = \frac{0+0}{1-0} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x}+3}{x-9}$$

Answer

$$0$$

4. (2 marks) Find the equation of the normal line to the ellipse $9x^2 + 4y^2 = 72$ at the point $(2, 3)$. (A normal line is perpendicular to a tangent line)

$$\frac{d}{dx}(\dots) = \frac{d}{dx}(72)$$

$$9(2x) + 4(2y) \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{9x}{4y}$$

Answer

$$y = \frac{2}{3}x + \frac{5}{3} \text{ or } 3y - 2x = 5$$

5. (2 marks) If a is a constant, evaluate $\frac{d^2}{dx^2}[\sin(ax)]$

$$\frac{d}{dx}(\sin ax) = a \cos ax$$

$$\frac{d}{dx}(a \cos ax) = a(-a \sin ax)$$

Answer

$$-a^2 \sin ax$$

Part C: Full Solution. Show all steps in your solution.

1. (10 marks) Given that f is differentiable at $x = 1$, $f(x) \neq 0$ for all $x \neq 0$ and $f(1) = 1$, evaluate the following:

$$\left. \frac{d}{dx} \left(\frac{\ln(x^2)}{f(x)} \right) \right|_{x=1}$$

$$\frac{d}{dx} \left(\frac{\ln(x^2)}{f(x)} \right)$$

$$= \frac{f(x) \frac{d}{dx} \ln(x^2) - f'(x) \ln(x^2)}{[f(x)]^2}$$

$$= \frac{f(x) \cdot \frac{1}{x^2} \cdot 2x - f'(x) \cdot 2 \ln x}{[f(x)]^2}$$

$$\text{at } x=1, \left. \frac{d}{dx} \left(\frac{\ln x^2}{f(x)} \right) \right|_{x=1} = \frac{f(1) \cdot 2 - f'(1) \cdot 2 \ln(1)}{[f(1)]^2}$$

$$\text{Since, } f'(1) \text{ exists, } = \frac{(1) \cdot 2 - f'(1) \cdot 0}{(1)^2} = 2$$

2. (10 marks) Differentiate the function $f(x) = \frac{1}{2+x}$, by forming the difference quotient:

$$\frac{f(x+h) - f(x)}{h}$$

and taking the limit as $h \rightarrow 0$

(give all relevant steps with justifications)

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{1}{2+x+h} - \frac{1}{2+x} \right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{\cancel{(2+x)} - \cancel{(2+x+h)}}{(2+x+h)(2+x)} \right)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{-h}{(2+x+h)(2+x)} \right)$$

$$= \lim_{h \rightarrow 0} \frac{-1}{(2+x+h)(2+x)}$$

$$= \frac{-1}{(2+x)(2+x)} = -\frac{1}{(2+x)^2}$$

3. (10 marks) Use squeeze theorem to show that $\lim_{x \rightarrow 0^+} \sqrt{x^2 + 2x} \sin\left(\frac{1}{\pi x}\right) = 0$
(give all relevant steps with justifications)

$$-1 \leq \sin\left(\frac{1}{\pi x}\right) \leq 1 \quad \text{for all } x \neq 0$$

$$\sqrt{x^2 + 2x} \geq 0 \quad \text{for } x \geq 0 \text{ and } x \leq -2$$

Multiplying the first inequality throughout:

$$-\sqrt{x^2 + 2x} \leq \sqrt{x^2 + 2x} \sin\left(\frac{1}{\pi x}\right) \leq \sqrt{x^2 + 2x} \quad \text{for all } x \geq 0, x \leq -2$$

$$\lim_{x \rightarrow 0^+} (-\sqrt{x^2 + 2x}) = 0$$

$$\lim_{x \rightarrow 0^+} (\sqrt{x^2 + 2x}) = 0$$

Therefore, by Squeeze theorem:

$$\lim_{x \rightarrow 0^+} \sqrt{x^2 + 2x} \sin\left(\frac{1}{\pi x}\right) = 0$$

4. (2 marks) BONUS QUESTION

By recognizing as the definition of derivative, evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{x} \right)$

(Warning: zero credit if L'hospital's rule is used.)

$$\text{let } f(x) = e^{2x} \quad f'(x) = 2e^{2x}$$

$$\lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{x} \right) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0) \\ = 2e^0 = 2$$